Experiment 10.1

Analysis of Critical Frequency and Maximum Usable Frequency (MUF) using MATLAB

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Aim

To analyse the critical frequency and the Maximum Usable Frequency (MUF) of the ionosphere for different electron densities and angles of incidence.

Software Used

MATLAB

Theory / Formulas

Critical frequency: $f_c = 9\sqrt{N_{max}}$, where N_{max} is the maximum electron density.
MUF = $f_c \sec(\theta)$, where θ is the angle of incidence.
MUF ratio = MUF / f_c = $\sec(\theta)$.

Procedure

1. Open MATLAB and create a new script.
2. Define the parameters and generate the signals/data.
3. Compute f_c for the range of N_{max}.
4. Compute the MUF and the MUF ratio for angles of incidence from 0 to 70 degrees.
5. Plot and compare f_c and MUF.

Output

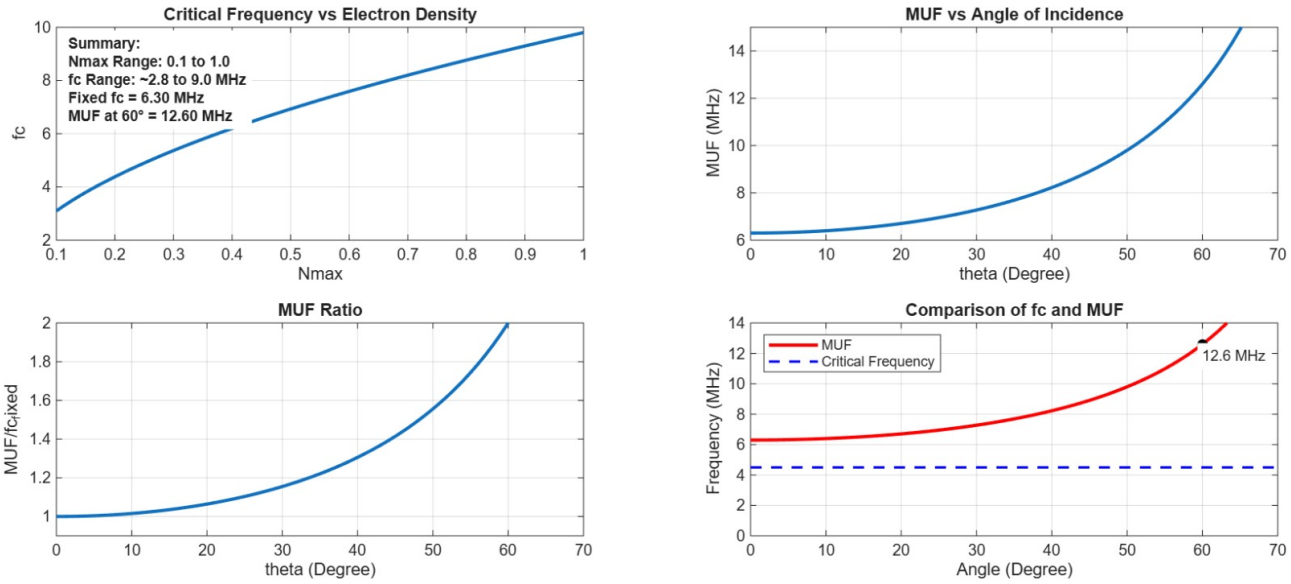


Fig. 1: Critical frequency vs electron density, MUF vs angle of incidence, MUF ratio, and comparison of f_c and MUF

Result

The critical frequency increases with electron density. For a fixed f_c of 6.30 MHz, the MUF at 60 degrees is 12.60 MHz, and the MUF increases with the angle of incidence.